		Change in BE = +179 MeV, manifested as the energy released (products are more stable)
30	**C**	For sample X: $A = \lambda_A N$ For sample Y: $3A = \lambda_Y N$ and since for half life $\lambda = \dfrac{\ln 2}{t_{\frac{1}{2}}}$ we can combine the equations as follows: $A = \dfrac{\ln 2}{t_{1/2(X)}} N$ $3A = \dfrac{\ln 2}{t_{1/2(Y)}} N$ $\dfrac{3A}{A} = \dfrac{\dfrac{\ln 2}{t_{1/2(Y)}} N}{\dfrac{\ln 2}{t_{1/2(X)}} N}$ $3 = \dfrac{t_{1/2(X)}}{t_{1/2(Y)}}$